From Detection Scores to Supported Claims: Measurement Audits for Document Analysis and Retrieval

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Abstract. Interpreting evaluation scores depends on what is observed, how it is scored and which comparisons the evidence supports. We audit a staged document evaluator, the released FinRank evidence-ranking benchmark and researcher-generated runs on two BEIR datasets. In the staged evaluator, 96 of 118 controlled header changes were visible. Under a single-responsive-leaf restriction, applying the same threshold to flat and hierarchical scores detected 75/95 versus 20/95 tables; a post-hoc scale-matched threshold recovered the same 75/95 tables on saved outputs. In FinRank, changing the tie policy shifted absolute pairwise accuracy while preserving the direction of the hard-versus-random comparison. For our BEIR runs, paired sign-flip tests with Holm correction supported two of ten lexical-system comparisons on SciFact and one on NFCorpus. An exact-tie-aware point-estimate baseline also passed our duplicate and degradation controls. These cases show how explicit measurement contracts qualify or preserve particular comparisons.

Keywords: measurement validity, document evaluation, information retrieval, reproducibility, benchmark auditing

1 Introduction

A lower detection rate can indicate a weaker evaluator, but it can also arise when the same numerical threshold is applied to scores on different scales. Likewise, a change in benchmark score need not change the comparison for which the score was reported; and the highest mean in a leaderboard does not by itself establish that every apparent ordering is supported by the data. Such distinctions matter when researchers use evaluation results to select systems or interpret a benchmark.

We ask: **which interpretations remain supported when the observation surface, scoring rule and comparison procedure are made explicit?** We investigate three evaluation constructions. First, controlled changes to public financial tables probe a staged document evaluator. Second, we inspect tie handling and source identity in FinRank [1]. Third, we assess researcher-generated lexical retrieval runs on SciFact and NFCorpus from BEIR [2], with a later dense-model exploration. The BEIR runs are our own constructions, not the published benchmark leaderboard.

Our contributions are three empirical findings. We explain a detection-rate difference by a positive score rescaling under a single-responsive-leaf restriction;

show that a particular FinRank comparison retains its direction despite changes in absolute scores; and distinguish point-estimate rankings from comparisons supported under a specified paired statistical procedure.

Prior work has examined metric and construct validity [3], [4], [5]. AgentSuite [6] audits interacting components in agent benchmarks at broader scale. We complement these perspectives with artifact-level analyses of observation, score scaling and comparison in document and retrieval evaluations.

Document analysis evaluation already addresses several related problems. SCORE [7] evaluates semantically equivalent but structurally different generative parsing outputs; Soric et al. [8] examine end-to-end and subtask table-extraction evaluation across heterogeneous PDFs. Those studies assess document parsing or extraction under their own evaluation designs. We instead audit the interpretation of measurements produced by a particular frozen evaluator, including what its selected mapping surface can observe and what follows from its scoring scale. Our table inputs are derived from financial QA data, not page-image OCR outputs. We do not benchmark document extraction models or claim a superior parsing metric.

The document case is the primary construction: a controlled change to a table header travels through a mapping rule into a checkpoint score and then a thresholded decision. Its observation and decision contracts can be inspected separately. The two external ranking cases complement this document-centered analysis by examining whether scores and comparison claims remain stable under other declared conventions; they do not turn the controlled header study into an external validation of a document-image analysis method.

2 Audit Design

We distinguish three questions: whether a change is **visible** at a selected evidence surface; whether a specified predicate **detects** it; and whether it changes an outcome under a stated equivalence contract. Visibility, detection and outcome change are separate observations. We then examine score sensitivity, the stability of a named comparison and numerical reproducibility.

The designs differ. Case 1 uses controlled perturbations. Case 2 inspects a released benchmark, including source-informed post-hoc identity analysis. Case 3 uses researcher-generated systems and test queries from a public benchmark; its original audit choices were not independently timestamped before the first run. We preserve fixed outputs and identify later explorations rather than treating them as prospective confirmation.

The three cases distinguish score differences from supported comparisons: a detection-rate gap need not imply a ranking gap, absolute accuracy changes need not reverse a comparison, and a higher mean need not establish every pairwise ordering. The following table summarizes these interpretations; each case specifies what is held fixed and what is varied.

For each construction we identify the input and outcome carrier, the implementation, the original reporting convention and the comparison of interest. We

Table 1. Interpretations qualified or retained by the three audits.

Case	Tempting interpretation	Supported interpretation
Document	Lower detection means weaker ranking	Rates differ; ranks coincide in the single-leaf setting.
FinRank	Absolute score changes imply a changed comparison	The random-minus-hard gap stays positive under both inspected tie policies.
Retrieval	A higher mean establishes every system ordering	The paired test supports some differences; other pairs are not proven equal.

use *decision* for an analytical interpretation such as a comparison's direction; no human-decision intervention was conducted. The cases use different units and populations, so their counts are not pooled into an overall audit-success rate.

3 Controlled Document Evaluation

Our project-built, frozen staged evaluator has parsing and question-answering checkpoint trees with twelve scored leaves. Its seed-42 panel contains 95 TAT-QA-derived source tables [9] and 118 controlled header changes across six families. Mapping and regression artifacts form the observed surface. For each injected header, the visibility analysis identifies its original canonical field: the change is counted as absorbed when that field still maps to the injected header, and visible otherwise. This matcher-relative criterion is not simply a character-level difference. Protocol detection and downstream harm are separate quantities. The audit holds the evaluated inputs and scores fixed while comparing declared aggregation and threshold conventions.

Of 118 injections, 96 were visible. None of the 13 OCR-like changes was visible on this surface. This observation does not establish 13 harmful misses: one existing semantic judge classified the cases as referentially equivalent, without an independent human reference. The family-level visible/injected counts were 43/47 alias variants, 15/17 date variants, 20/20 mixed-language variants, 0/13 OCR-like variants, 13/16 prefix-noise variants and 5/5 unit-suffix variants. These are injection counts in the seed-42 panel; the 118 changes are clustered within 95 tables. They characterize the particular matcher-relative surface, not generic document defects or actual OCR error rates.

In this construction the QA tree was frozen and only the field-hit-rate leaf responded. For field hit rate h, the aggregation weights give a parsing-root score of approximately $6.25 + 3.75h$, while the flat drop is ten times the hit-rate drop. The hierarchical drop is therefore approximately 0.375 times the flat drop, and their observed rankings coincide. The original protocols each used a numerical operating threshold of **at least 2.0** on their respective scales. At those thresholds the flat and hierarchical protocols detected 75/95 versus 20/95 tables, or 85/118

versus 13/118 injections. Tables and injections are distinct units; a rate difference does not by itself establish a ranking difference.

Table 2. Seed-42 original thresholds and post-hoc scale matching; table and injection counts are distinct units.

Unit	Flat ≥ 2	Tree ≥ 2	Tree ≥ 0.75
Source tables	75/95	20/95	75/95
Header injections	85/118	13/118	85/118

The flat threshold of 2.0 corresponds to 0.75 on the hierarchical scale. In a post-hoc check of saved rounded scores, these matched thresholds select the same 75/95 tables and 85/118 injections as the flat rule. A separate grid used strict greater-than thresholds of 1.0 and 0.375: in one recorded fold both selected the same 21 of 24 injections with experimental H/HI labels. The predicates must not be conflated: 12 of the 95 tables have a flat drop exactly equal to 2.0.

In this single-leaf setting, a positive rescaling preserves score order; retaining the same decision rule also requires rescaling its threshold. The matched-grid example uses saved outputs and does not calibrate a deployment threshold or establish equivalence with multiple responsive leaves.

The clean-only contrastive construction compares repeated deterministic computations on an unchanged record. All 2,382 usable clean records have zero scores, so every positive threshold has observed clean false-positive rate zero; this panel does not identify a unique deployment threshold. Across three seeds, the flat descriptive table-run ratio was 219/282 using 119 unique source-table identifiers. A post-hoc UID-cluster bootstrap gives a 95% interval of [0.7117, 0.8364]; the repeated table runs are not 282 independent source tables.

The per-seed contributions to the 219/282 ratio are 75/95, 76/97 and 68/90. The denominators refer to table appearances within each run; the same source table may contribute to more than one run. The bootstrap samples source identifiers with replacement, includes all run appearances for each sampled identifier, and recalculates the pooled ratio. This uncertainty interval is post-hoc and conditional on the available source tables and three seeds; it should not be interpreted as a guarantee for future datasets or seeds. Equal weighting of source identifiers gives 0.7535 with interval [0.6835, 0.8207].

In a separate stage-aligned control, 72 source clusters contributed one injection each for mapping, retrieval and numeric processing. A leave-one-cluster-out nearest-centroid classifier on the full response vector attained macro-F1 and top-1 1.000. The three directly targeted parsing, retrieval and reconciliation leaves gave the same result; removing them reduced top-1 to 0.333 under the frozen tie-as-incorrect rule, with a 0.667 tie rate. The classifier was evaluated by leaving one source cluster out at a time, so the three injected stages from the held-out cluster were not included when fitting its centroids. This ablation tests the source of the observed signal in the controlled instrument, not error localization on an independent deployed system. The controlled stage signal thus resides in

the directly targeted leaves. A nested FinQA probe [10] covered 89 rows from 30 report clusters; it changed program fields but did not recompute downstream result fields. Its zero detected and zero changed outcomes cannot estimate a failure rate for changed answers.

4 FinRank: Score Sensitivity and Decision Stability

FinRank ranks evidence from financial filings [1]. Under tie-as-positive scoring, a score tie counts for the gold passage; under strict-win scoring, it does not. The tie comparisons and materiality threshold were fixed before pair-level outcomes. The following table shows that random-negative accuracy moves by 1.084 percentage points for TF-IDF and 1.029 for BM25, exceeding the fixed 1.0-point threshold, while the random-minus-hard gap remains positive in both systems.

Table 3. FinRank pairwise accuracy under two tie policies; gaps are computed before rounding.

System	Policy	Hard (%)	Random (%)	Gap (pp)
TF-IDF	Positive	77.756	91.935	14.179
TF-IDF	Strict	77.030	90.850	13.821
BM25	Positive	74.780	87.789	13.009
BM25	Strict	74.094	86.760	12.666

Gaps are random accuracy minus hard accuracy, computed before display rounding. The gap's direction and descriptive TF-IDF/BM25 ordering remain stable under both policies. A source-informed post-hoc analysis of the 1,185-record release reproduces 442 hard-negative occurrences across 305 records sharing canonical text with another record's gold passage. The inspected pooled corpus keeps the first occurrence's metadata when deduplicating passage text; metadata filtering can consequently exclude a later record's gold text, as observed for 33 records. These properties were known from the source. In-record and pairwise metrics reproduce exactly in the audited environment; global and metadata-filtered metrics differ from the released results by up to 0.19 and 0.16 percentage points, respectively. The cause remains unresolved.

5 Retrieval Comparisons

We built five lexical configurations on BEIR SciFact and NFCorpus [2]: TF-IDF and BM25 parameter pairs (k_1, b) of (0.9, 0.4), (1.2, 0.75), (0.5, 0.3) and (1.5, 0.6). We concatenate each document's title and text, lowercase and tokenize on ASCII alphanumeric runs, use log-scaled term frequency and smoothed inverse document frequency for TF-IDF cosine, and resolve equal scores by ascending document ID unless testing the alternative order. They use the released corpora, qrels and 300/323 judged queries, respectively. The systems are our own runs,

and the two datasets belong to one benchmark family. We compare system scores on the same queries so that statistical analysis is paired; each dataset has its own comparison family.

Our initial scoring implementation binarized graded relevance and used a different MAP denominator. Aligning it with the trec_eval contract corrected both definitions. Under graded nDCG@10, TF-IDF moved from second to third in one NFCorpus ranking; the separate denominator correction changed TF-IDF's MAP@10 from approximately 0.2200 to 0.1162. This is a correction of our implementation, not evidence that the official leaderboard was wrong. For fixed rankings, trec_eval 10.0-rc3 agreed with 18,690 per-query values within 10^{-4}: five systems, 623 queries and six nDCG, MAP and recall cutoff combinations. This check does not validate tie-breaking policy.

The cross-check encodes each fixed rank with a strictly decreasing score. Its maximum discrepancy was approximately 0.00005, consistent with the reference program's four-decimal output. It checks metric arithmetic for the chosen ranking, rather than which document should win a score tie; the executable was trec_eval itself, not the pytrec_eval Python binding.

For the five systems, 2,000 paired bootstrap resamples produced intervals excluding zero for 3/10 pairs on SciFact and 4/10 on NFCorpus. A separate two-sided paired sign-flip procedure with 2,000 Monte Carlo draws per pair and Holm correction supported 2/10 and 1/10, respectively. Both sign-flip analyses use the Monte Carlo correction $(b+1)/(M+1)$, where b is the number of draws at least as extreme as observed and M the number of draws. Bootstrap and sign-flip are different inference procedures, so the count difference is not attributable to Holm correction alone. Inference depends on the relevant exchangeability or symmetry and query-independence assumptions; non-significance does not establish equality.

The supported SciFact pairs were BM25(1.5, 0.6) versus TF-IDF (mean nDCG@10 difference +0.03334; Holm-adjusted $p = 0.0100$) and BM25(1.2, 0.75) versus TF-IDF (+0.02930; $p = 0.0450$). On NFCorpus, BM25(1.5, 0.6) versus BM25(0.5, 0.3) had a difference of +0.00817 ($p = 0.0300$). A later 200,000-draw sensitivity check on regenerated lexical query scores retains the same three supported pairs when Holm is recalculated within the five-system family. It reduces Monte Carlo variability but does not independently validate the data or inferential assumptions.

Reversing document-ID tie order for BM25(0.9, 0.4) on NFCorpus at k=10 changed the ordered top ten for 36.5% of queries and top-ten membership for 25.7%. Mean nDCG@10 changed from 0.30453945 to 0.30449093, approximately 0.00005. Query-level changes can therefore be hidden by the aggregate in this setting; this is not an estimated benchmark-wide failure rate.

For an intentionally duplicated system and an intentionally degraded control, an exact-tie-aware point-estimate rule and the paired audit both passed both controls. The controls do not demonstrate that the audit outperforms the simpler rule. A subsequent BGE-small-en-v1.5 [11] exploration gave mean nDCG@10 of 0.7127 versus 0.6622 for BM25(1.2, 0.75) on SciFact, and 0.3438 versus 0.3062

on NFCorpus. This is an exploratory retrieval result selected after the original analysis, not evidence that the audit improved a user decision or that lexical tie findings transfer to dense models.

The BGE comparison was later included with the five lexical runs in a separate six-system family of 15 pairs per dataset. With 200,000 Monte Carlo sign flips per pair and Holm correction within each dataset, the BGE-minus-BM25(1.2, 0.75) nDCG@10 difference was +0.05049 on SciFact (unadjusted $p = 0.0064$; adjusted $p = 0.0660$) and +0.03759 on NFCorpus (adjusted $p = 0.0155$). The SciFact pair crosses the 0.05 threshold after adjustment within this same test; it is not evidence of equality. These post-hoc tests are not the original ten-pair family or independent confirmation. The saved dense outputs contain ranked document IDs rather than original dense scores, so they cannot establish dense-score tie frequencies. Numerical reproducibility of the tests does not establish their inferential assumptions.

The earlier single-pair BGE analysis used normalized embeddings, the model's query prefix and a 512-token cap. Its uncorrected paired-bootstrap differences were 0.0505 [0.0137, 0.0891] on SciFact and 0.0376 [0.0154, 0.0608] on NFCorpus. A separate trec_eval run agreed on 1,246 dense and BM25 per-query scores to four-decimal print precision. The later 15-pair test addresses multiplicity, not prospective confirmation.

6 Discussion and Conclusion

The three cases show distinct ways in which an evaluation contract constrains a claim. In the staged evaluator, an apparent rate difference vanishes after matching thresholds under a restricted score rescaling. In FinRank, a tested comparison retains direction as absolute scores change. In our BEIR runs, tie handling and statistical comparison procedures affect the interpretation of rankings; correction of our own metric implementation is a separate calibration step.

These findings have limited scope: the first case uses controlled perturbations to structured tables, without a page-image recognition experiment or an independent human reference for the OCR-like variants. FinRank uses lexical baselines and has an unresolved reproduction gap, and the BEIR analyses are retrospective runs over two datasets with a post-hoc dense extension. Existing validity frameworks already cover broader audit principles [3], [4], [5]. Our contribution is a reproducible set of case-specific observations and the distinction between numerical movement and supported interpretation. We recommend reporting the measurement object, scoring convention, comparison and uncertainty together; we do not claim a general audit guarantee or observed deployment benefit.

7 Data and Materials

The TAT-QA, FinRank and BEIR data used in this study are described in [9], [1] and [2], respectively; the BEIR runs use released relevance judgments. Study-specific audit scripts, configuration records and aggregate results are retained as

research artifacts. Access to and redistribution of underlying data remain subject to the respective source terms.

Acknowledgment

Generative AI tools, including DeepSeek [12] and OpenAI Codex [13], assisted with analysis-code development and drafting and revision of manuscript text in Sections 1–6. Google Gemini [14] provided review suggestions on earlier drafts. The authors are responsible for the final manuscript and its conclusions.

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